American Computer Science League

2022-2023 ● Contest 1: Short Problems Solutions ● Senior Division

1. Computer Number Systems

Decimal	Octal	Decimal	Octal	Decimal	Octal
100	144	110	156	120	170
101	145	111	157	121	171
102	146	112	160	22	172
103	147	113	161	123	173
104	150	114	162	124	174
105	151	115	163	125	175
106	152	116	164	126	176
107	153	117	165	127	177
108	154	118	166	128	200
109	155	119	167	129	201

There are 12 occurrences of each digit 5, 6, or 7 for a total of 36.

A. 36

2. Computer Number Systems

$FA8072_{16} - 40E0D0_{16} = B99FA2_{16}$ by borrowing 16 if needed.

$B99FA2_{16} = 101110011001111110100010_2$
$= 101\ 110\ 011\ 001\ 111\ 110\ 100\ 010_2 = 56317642_8$

C. 56317642

3. Recursive Functions

$f(17) = f(17-5)+17 = f(12)+17 = 18+17 = 35$
$f(12) = f(12-5)+12 = f(7)+12 = 6+12 = 18$
$f(7) = f(7-5)+7 = f(2)+7 = -1+7 = 6$
$f(2) = f(2+2)-2 = f(4)-2 = 1-2 = -1$
$f(4) = f(4+2)-4 = f(6)-4 = 5-4 = 1$
$f(6) = f(6-5)+6 = f(1)+6 = -1+6 = 5$
$f(1) = -1$

D. 35

4. Recursive Functions

$f(12,18) = f(12/2, 18-3) - 12 * 18 = f(6,15) - 216$
$= -115 - 216 = -331$
$f(6,15) = f(6/2, 15-3) - 6 * 15 = f(3,12) - 90$
$= -25 - 90 = -115$
$f(3,12) = f(12+2, 3/3) + 3 = f(14,1) + 35$
$= -28 + 3 = -2$
$f(14,1) = f(14/2, 1-3) - 14 = f(7,-2) - 14$
$= -14 - 14 = -28$
$f(7,-2) = 7 * -2 = -14$

A. -331

5. What Does This Program Do?

The outer loop checks the numbers 3, 6, 9, 12, 15, and 18. The inner loop counts the number of powers of 2 that are less than that number. Therefore, for 3, it is 2; for 6, it is 2, 4; for 9, it is 2, 4, 8; for 12, it is 2, 4, 8; for 15, it is 2, 4, 8; and for 18, it is 2, 4, 8, 16. That's a total of 16.

E. 16